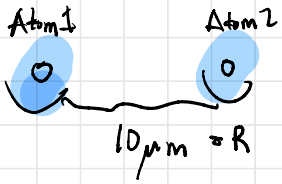
Rydberg atoms:

- Dipole moment → $e \cdot r \propto n^2$



Dipole-dipole:

$$V_{rr} = \left(\frac{n^2 e a_0}{R^3}\right)^2 \longrightarrow \propto n^4$$

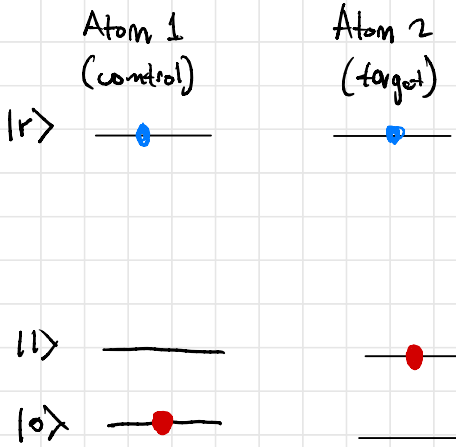
$\sim$ 250 MHz

Compare to Γ (natural linewidth)

$\sim$ 6 MHz.

Two atom gates:

1. Traps
2. Light pulses. → optical (two photon 780 & 480 nm)



Focus on:

$$|01\rangle \; ; \; |10\rangle \; ; \; |00\rangle \; ; \; |11\rangle .$$

(3) Pulse sequence. → all pulses are resonant w/ the unperturbed Rydberg transition.

Pulses:

1. π-pulse on control
2. 2π-pulse on target
3. π pulse on control

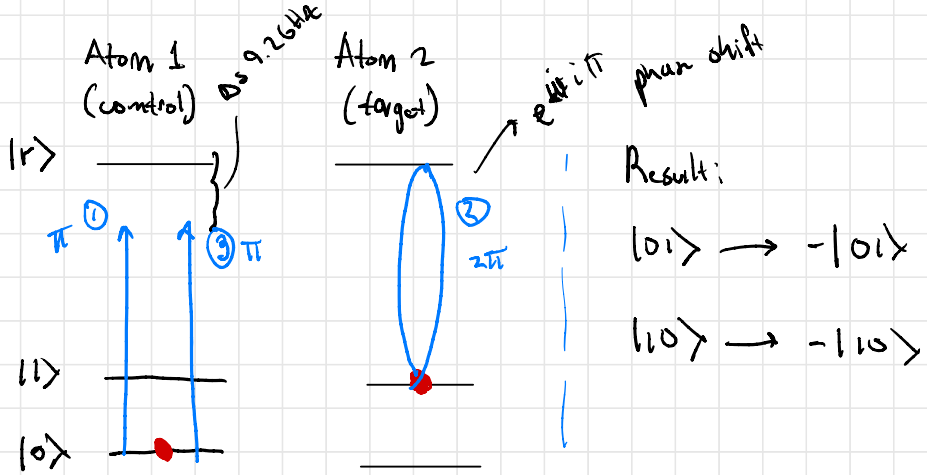
} Note: phase coherent. (rotations around X-axis)

All resonant on $|1\rangle \rightarrow |r\rangle$.

Case 1:

$|01\rangle$

or $|10\rangle$



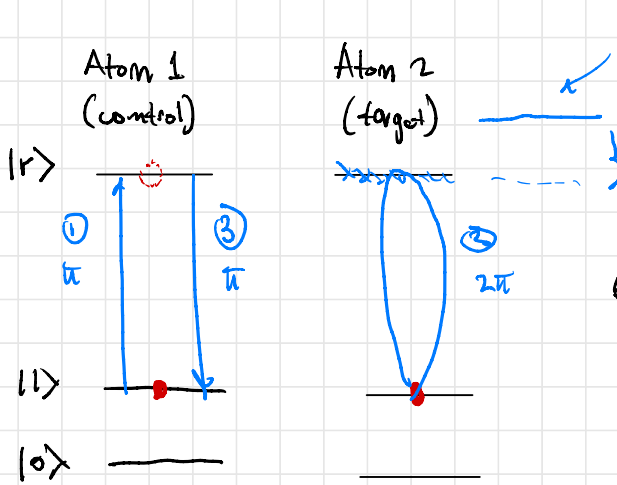
Result:

$|01\rangle \rightarrow -|01\rangle$

$|10\rangle \rightarrow -|10\rangle$

Case 2:

$|11\rangle$



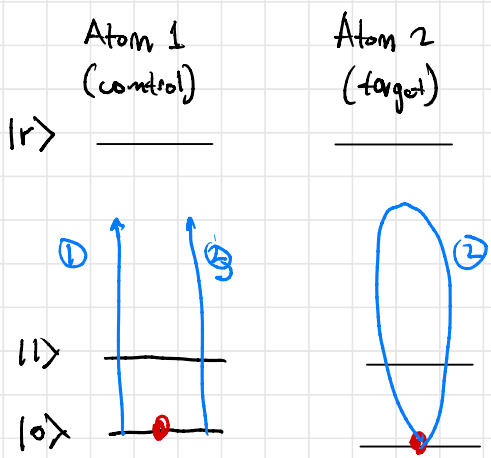
$\Delta\varphi_t = 0$

$\Delta\varphi_c = \pi$

$|11\rangle \rightarrow -|11\rangle$

Case 3:

$|00\rangle$

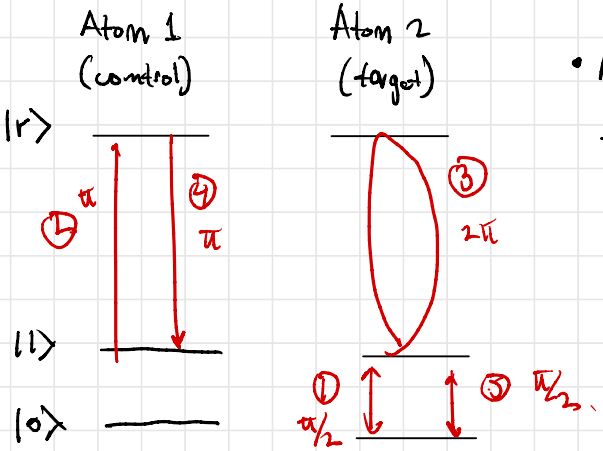


$|00\rangle \rightarrow |00\rangle$

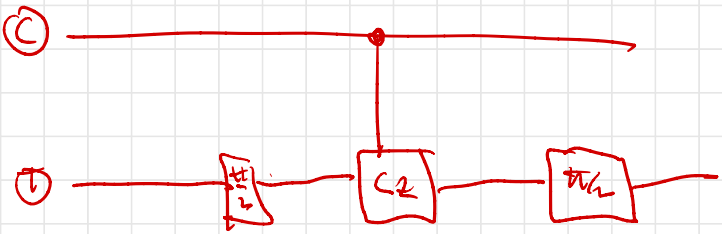
$$\begin{array}{c} \\ \langle 00| \\ \\ \langle 11| \end{array}\begin{array}{c} \begin{array}{cccc} |00\rangle & & & |11\rangle \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \end{array} \simeq -1 \otimes \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

"Rydberg CZ" ⟶ Entangling gate.

Getting a CNOT gate.

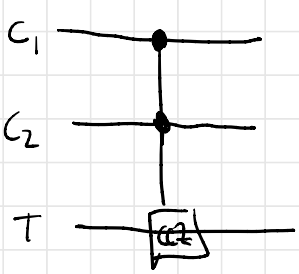


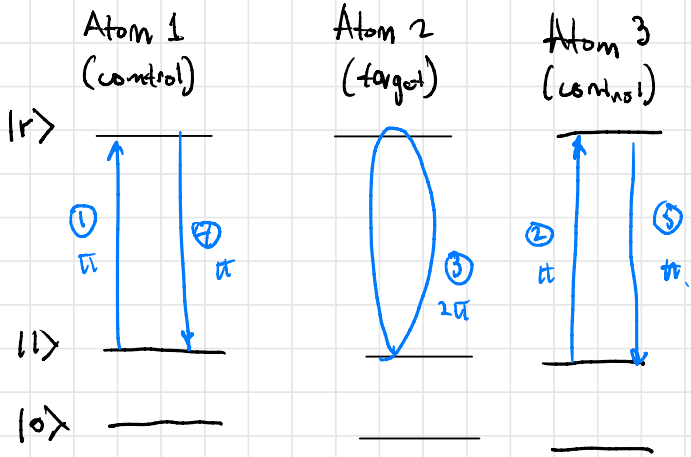
- Apply a (A) to target before and after the CZ.



Multi qubit gates: i.e. CCZ gate.

$$\begin{pmatrix} -1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{pmatrix}$$





Performance:

Best fidelity for 2-qubit ~ 99.84 %
(MIT – Harvard/Quera)
post selected on atom loss

→ getting to 99.9 % is the current goal.